Class 7

Physics

Volume – 2

Solutions

Table of Content

6.	Work Power And Energy	1 – 20
7.	Heat	21 – 24
8.	Circular Motion	25 – 35
9.	Optics	36 – 42

6.WORK POWER AND ENERGY

WORKSHEET – 1

LEVEL – I

1.A	2. B	3. C	4. B	5. D	6. A
7. B	8. D	9. C	10. A		

LEVEL – II

1.A	2. A	3. C	4. A	5. B	6. A
7. A	8. B	9. A	10. A	11. C	12. C
13. C	14. B	15. C	16. A	17. C	18. A
19. B	20. A-S; B-R; C-Q; D-P		21. A, B, C, C		22. A-Q, B-P, C-R, D-S, T
23. C	24. A	25. B			

SOLUTIONS

LEVEL – II

2. $\text{m} = 80\ \text{kg} + 20\ \text{kg}$
   $\text{m} = 100\ \text{kg}$
   $\text{h} = 40 \times 25\ \text{cm} = 1000\ \text{cm} = 100\ \text{m}$
   $\text{h} = 10\ \text{m}$
   $g = 9.8$
   $\text{w} = \text{mgh}$
   $\text{w} = 100 \times 9.8 \times 10$
   $\text{w} = 9800\ \text{J}$

5. $m = \frac{1}{10} g$
   $m = \frac{1}{10} \times \frac{1}{1000}\ \text{kg}$
   $h = 100m$
   $g = 9.8 m/s^2$
   $w = mgh\ w = \frac{1}{10000} \times 9.8 \times 100$
   $w = 0.098J$

6. $m = 70\ \text{kg}$
   $h = 36 \times 20\ \text{cm} = 36 \frac{20}{100}\ \text{m}$
   $g = 9.8\ \text{m/s}^2$
   $w = mgh$
   $w = 70 \times 9.8 \times 36 \times \frac{20}{100}$
   $w = 4939.2J$

7. $F = 5N$
   $\text{s} = 20\ \text{cm} \frac{20}{100}\ \text{m}$
   $w = F \times s$
   $w = F \times s$
   $w = 1J$

9. $m = 5$ kg
$h = 400$ cm
$g = 9.8$ m/s^2
$w = mgh$
$w = 5 \times 9.8 \times \frac{400}{100}$
$w = 196J$

10. $F = 5N$
$s = 5m$
$w = F \times S$
$w = 5 \times 5$
$w = 25J$

11. $m = 50$ kg
$F = 100N$
$\theta = 60°$
$s = 10m$
$w = FS$
$w = 50 \times 10$
$w = 500J$

15. $F = 2.50$ N
$m = 15g$
$s = 2.5m$
$w = F.s$
$w = 2.5 \times 2.5$
$w = 6.25J$
$PE = 1J$
$h = ?$
$g = 9.8$ m/s^2
$PE = mgh$
$h = \frac{PE}{mg}$
$h = \frac{1}{1 \times 9.8}$
$h = 0.102m$

16. $m = 55$ kg
$h = 0.15m$ ($\because$ h = 2R)
$w = mgh$
$w = 55 \times 9.8 \times 0.15$
$w = 3234J$

17. $h = 5m$
$g = 10m/s^2$
$w = 2500J$
$w = mgh$
$m = \frac{w}{gh}$
$m = \frac{2500}{10 \times 5}$
$m = 50$ kg

18. $m = 200$ kg
$h = 10m$
$g = 9.8m/s^2$

$w = mgh$
$w = 200 \times 9.8 \times 10 = 19600J$

19. Given, w = 735 J
h = 5 m
g = 9.8 m/s^2
$W = mgh \Rightarrow m = \frac{w}{gh}$

$= \frac{735}{5 \times 9.8} = 15$ kg

23. $m = 1$ kg
$u = 9.9$ m/s
$t = 1$sec
$s = ut - \frac{1}{2}gt^2$
$s = 9.9(1) - \frac{1}{2}(9.8)(1)^2$
$s = 5m$
$w = -mgs$
$w = -1 \times 9.8 \times 5$
$w = -49J$

24. $w = mgh$
$w = 2 \times 9.8 \times 1$
$w = 19.6J$

25. $m = 15$ kg
$s = 1.5m$
w = mgh = 15 × 10 × 1.5 = 225 J
$w = 225J$

WORKSHEET – 2

LEVEL – I

1.C	2. C	3. D	4. B	5. C	6. A
7. B	8. B	9. D	10. C		

LEVEL – II

1. B	2. C	3. A	4. A	5. D	6. B
7. B	8. C	9. A	10. A	11. A	12. A
13. A	14. C				

LEVEL – III

15. A, B, C, D	16. B	17. B	18. B	19. A-P, B- Q, C-R, D-T,S
20. C	21. A	22. B	23. A-R, B-P, C-Q, D-S	
24. A	25. C	26. D	27. A-S, B–R, C-Q, D-P	

SOLUTIONS

LEVEL – II

1. $w = 1000J$
$t = 10$sec
$p = \frac{w}{t}$ $p = \frac{1000}{10}$
p = 100 W

2. $\text{p} = 1500\ \text{W}$
$t = 1hr$
$t = 3600\text{sec}$
$p = \frac{w}{t}$
$w = p \times t$
$w = 1500 \times 3600$
$w = 54 \times 10^5\ \text{J}$

4. $F = 600\ \text{N}$
$v = 20\ \text{m/s}$
$p = F \times v$
$p = 600 \times 20$
$\text{p} = 12000\ \text{W}$
$\text{p} = 12\ \text{kW}$

5. $w = 50J$
$\text{p} = 20\ \text{W}$
$p = \frac{w}{t}$
$t = \frac{w}{p}$
$t = \frac{50}{20}$
$t = 2.5\text{sec}$

6. $w = 180\ \text{J}$
$t = 1\ \text{min} = 60\text{sec}$
$p = \frac{w}{t}$
$\text{p} = \frac{180}{60} = 3\ \text{W}$

8. $t = 1\text{sec}$
$v = 3m/s$
$m = 40\ \text{kg}$
$p = \frac{w}{t}$
$p = \frac{1/2mv^2}{t}$
$p = \frac{1/2 \times 40 \times 3^2}{1}$
$p = 20 \times 9$
$p = 180w$

9. $\text{p} = 2\ \text{kW}$
$p = 2 \times 1000\ \text{W}$
$h = 10m$
$g = 10m/s^2$
$t = 1\text{sec}$
$p = \frac{w}{t}$
$p = \frac{mgh}{t}$
$2000 = \frac{m \times 10 \times 10}{60}$
$m = 1200\ \text{kg}$

10. $m = 100$ kg
$h = 100m$
$t = 10$sec
$p = \frac{w}{t}$
$p = \frac{mgh}{t}$
$p = \frac{100 \times 9.8 \times 100}{10}$
$p = 9800$ W
$p = 9.8$kw

11. Total force = ma - resistance force
F $= 400 - 50$
F $= 350$
$p = F \times v$
$p = 350 \times 10$
P $= 3500$w
$p = 3.5$ kW

12. We know that,
$\eta\gamma \cdot$ p $= \frac{\text{W}}{\text{t}}$ [$\because \eta =$ efficiency]
$v = 30m^3$
$h = 60m$
$\eta\% = 30\%$
$t = 10$ min
$\eta\% p = \frac{w}{t}$
$\eta\% p = \frac{mgh}{t}$
$\frac{30}{100} p = \frac{d \times v \times g \times h}{t}$
$\frac{30}{100} p = \frac{1000 \times 30 \times 10 \times 60}{10 \times 60}$
$p = 100000$ W
$p = 100$ kW

13. p $= 200$ W
$t = 1$min40sec
$t = 100$sec
$p = \frac{\text{energy consumed}}{\text{time}}$
$E = p \times t$
$E = 200 \times 100$
$E = 20000J$
60% of energy is used
$E = 20000 \times \frac{60}{100}$
$E = 12000J$

14. F $= 600 + 150$ N
F $= 750$ N
T $= 15$sec
s $= 6$ m

$p = \frac{w}{t}$

$p = \frac{F \times s}{t}$

$p = \frac{750 \times 6}{15}$

$p = 300W$

LEVEL – III

16. $p = \frac{w}{t}$

$p\alpha \frac{1}{t}$

So boy delivers more power

17. $p\alpha \frac{1}{t}$

$\uparrow P_A = \frac{1}{10 \downarrow}$

$\downarrow P_B = \frac{1}{20 \uparrow}$

$p_1 > p_2$

18. p = 11 W

$w = 440J$

$t = \frac{w}{p}\ \left(\because \mathrm{p} = \frac{\mathrm{W}}{\mathrm{t}}\right)$

$t = \frac{440}{11}$

$t = 40\text{sec}$

20, 21. $m = 50$ kg

$h = 40 \times 20 \times 10^{-2}$

$g = 10\ \mathrm{m/s^2}$

$p = \frac{w}{t}$

$p = \frac{mgh}{t}$

$p = \frac{50 \times 10 \times 40 \times 20 \times 10^{-2}}{50}$

p = 80 W

$\therefore$ W = p × t = 80 × 50 = 4000 J

22. Power in kW is $\rightarrow p = 80 \times \frac{1}{1000}$ kW $p = 0.08kW$

23. $m = 10 \times 60$ kg = 600 kg

$h = 20m$

$t = 30\text{sec}$

$g = 10m/s^2$

$p = \frac{w}{t}$

$p = \frac{mgh}{t}$

$p = \frac{600 \times 10 \times 20}{30} W$

$p = \frac{600 \times 10 \times 20}{30} \times \frac{1}{750}\left(\because 1\ \mathrm{W} = \frac{1}{750}\mathrm{hp}\right)$

$p = 5.33hp$

25. $v = 2000m^3$
$h = 20m$
$t = 25 \text{ min} = 25 \times 60\text{sec}$
$p = \frac{w}{t}$
$p = \frac{d \times v \times g \times h}{t}$
$p = \frac{1000 \times 2000 \times 10 \times 20}{25 \times 60} W$
$p = \frac{1000 \times 2000 \times 10 \times 20}{25 \times 60} \times \frac{1}{750}$
$p = 355.55\text{hp}$

26. $p = \frac{w}{t}$
$p = 2 \times 750 \text{ W}$
$V = 5m^3$
$h = 15m$
$t = \frac{mgh}{p}$
$t = \frac{1000 \times 5 \times 10 \times 15}{2 \times 750}$
$t = 500\text{sec}$

WORKSHEET – 3

LEVEL – I

1. B 2. B 3. B 4. A 5. C 6. C
7. C 8. C

LEVEL – II

1. B 2. A 3. A 4. C 5. A 6. D
7. D 8. C 9. A 10. A 11. D 12. B
13. A 14. B 15. A

LEVEL – III

16. A, B 17.B 18. B 19. A 20. A-R, B-S, C-P,Q, D-R
21. A, B, C, D 22. A-S, B-Q, C-P,D-R 23. A, C, D 24. C 25. C 26. D

SOLUTIONS

LEVEL – II

1. $m = 1000 \text{ kg}$
$KE = 20{,}000J$
$KE = \frac{1}{2}mv^2$
$v = \sqrt{\frac{2 \times KE}{m}}$
$v = \sqrt{\frac{2 \times 20{,}000}{1000}}$
$v = \sqrt{40}$

$v = 2\sqrt{10}$ m/s

2. $m = 2$ kg
$v = 20$ m/s
$KE = \frac{1}{2}mv^2$
$KE = \frac{1}{2} \times 2 \times 20^2$
$KE = 400J$

3. $KE = 625$ J
$m = 50$ kg
$KE = \frac{1}{2}mv^2$
$625 = \frac{1}{2} \times 50 \times v^2$
$v^2 = 25$
$v = 5$ m/s

4. $m = 10$ kg
$v = 2$ m/s
$KE = \frac{1}{2} \times mv^2$
$KE = \frac{1}{2} \times 10 \times 4$
$KE = 20J$

5. $v = 54$ km/hr
$v = 54 \times \frac{5}{18}$ m/s
$v = 15$ m/s
$KE = \frac{1}{2}mv^2$
$KE = \frac{1}{2} \times 40 \times 15^2$
$KE = 20 \times 225$
$KE = 4500J$

6. $v_1 = v$
$v_2 = 2v$
$\frac{KE_1}{KE_2} = \frac{1/2mv_1^2}{1/2mv_2^2}$
$\frac{KE_1}{KE_2} = \frac{1/2m(v^2)}{1/2m4v^2}$
$KE_1 : KE_2 = 1:4$

7. $m = 120g$
$m = 120 \times \frac{1}{1000}$ kg
$m = 0.12$ kg
$v = \sqrt{2^2 + 5^2}$
$v^2 = 29$
$KE = \frac{1}{2} \times 0.12 \times 29$
$KE = \frac{3.48}{2}$
$KE = 1.74J$

9. $F = 8N$
$m = 2kg$
$t = 10\text{sec}$
$u = 0$ m/s
$F = m\left(\frac{v-u}{t}\right)$
$8 = 2\left(\frac{v-0}{10}\right)$
$v = 40$ m/s
$KE = \frac{1}{2}mv^2$
$KE = \frac{1}{2} \times 2 \times 1600$
$KE = 1600J$

10. $d_1 = 1; v_1 = 2$
$d_2 = 2; V_2 = 1$
$r_1 = 1$
$r_2 = 2$
$KE = \frac{1}{2}mv^2$
(= density × volume)
$\frac{KE_1}{KE_2} = \frac{d_1 \times v_1 \times v_1^2}{d_2 \times v_2 \times v_2^2}$
$\frac{KE_1}{KE_2} = \frac{1 \times \frac{4}{3}\pi r_1^3 \times v_1^2}{2 \times \frac{4}{3}\pi r_2^3 \times v_2^2}$
$\frac{KE_1}{KE_2} = \frac{1}{2} \times \frac{(1)^3}{(2)^3} \times \frac{2^2}{1^2}$
$\frac{KE_1}{KE_2} = \frac{1}{2} \times \frac{1}{8} \times \frac{4}{1}$
$\frac{KE_1}{KE_2} = \frac{1}{4}$

11. $KE_1 = \frac{1}{2}mv^2 \rightarrow (1)$
$KE = \frac{1}{2} \times \frac{m}{2} \times (2v)^2$
$KE_1 = \frac{1}{2} \times \frac{m}{2} \times 4v^2 \rightarrow (2)$ $\frac{KE_1}{KE_2} = \frac{\frac{1}{2}mv^2}{\frac{1m}{2\,2} \times 4v^2} \times 100\%$
$KE_1 = 100\% KE_2$

12. $\frac{1/2mv^2}{1/2m(2)^2} = \frac{KE}{4KE}$
$\frac{v^2}{4} = \frac{1}{4}$
$v^2 = 1$
$v = \sqrt{1}$
$v = 1$ m/s

13. $m_1 = 3$
$m_2 = 1$
$E_1 = 3$
$E_2 = 1$

$KE = \frac{1}{2}mv^2$

$2KE = mv^2$

$v^2 = \frac{2KE}{m}$

$v = \sqrt{\frac{2 \times E}{m}}$

$\frac{v_1}{v_2} = \sqrt{\frac{\frac{2 \times 3}{3}}{\frac{2 \times 1}{1}}}$

$\frac{v_1}{v_2} = \sqrt{\frac{1}{1}}$

$v_1 : v_2 = 1:1$

14. $KE = \frac{1}{2}mv^2$

$KE = \frac{1}{2} \times 200 \times 5^2$

$KE = 2500\text{erg}$

15. $w = F.S$

$\frac{1}{2}mu^2 = F \times s \rightarrow (1)$

$m_1 = m + \left(\frac{40}{100}\right)m$

$m_1 = 1.4m$

$\frac{1}{2}m_1u^2 = Fs_1$

$\frac{1}{2}1.4mu^2 = F \times s_1 \rightarrow$

Equation (1) & (2)

$\frac{\frac{1}{2}mu^2}{\frac{1}{2}1.4mu^2} = \frac{F \times s}{F \times s_1}$

$s_1 = 1.4s$

LEVEL – III

17. $m = 60$ kg

$KE = 750J$

$E = \frac{1}{2}mv^2$

$v = \sqrt{\frac{2 \times E}{m}}$

$v = \sqrt{\frac{2 \times 750}{60}}$

$v = 5$ m/s

18. $E = 5J$

$v = 2$ m/s

$E = \frac{1}{2}mv^2$
$5 = \frac{1}{2}m(2)^2$
$10 = 4m$
$m = 2.5$ kg

19. $KE = \frac{1}{2}mv^2$
$KE = \frac{1}{2} \times 400 \times (20)^2$
$KE = 200 \times 400$
$KE = 80000$erg

24. $m = 4$ kg
$g = 10m/s^2$
$t = 5$sec
$v = u + at$
$v = 0 + 5(10)$
$v = 50$ m/s
$KE = \frac{1}{2}mv^2$
$KE = \frac{1}{2} \times 4 \times (50)^2$
$KE = 5000J$
$KE = 5KJ$

25. $h = 0.2m$
$v = \sqrt{2gh}$
$v = \sqrt{2 \times 0.2 \times 9.8}$
$v = 1.4$ m/s

26. $m = 0.1$ kg
$h = 10m$
$KE = \frac{1}{2}mv^2$
$v = \sqrt{2gh}$
$v = \sqrt{2 \times 10 \times 10}$
$v = 14.14$
$KE = \frac{1}{2}mv^2$
$KE = \frac{1}{2} \times 0.1 \times 14.14 \times 14.14$
$KE = 19.99 \approx 20$
$KE = \frac{1}{2} \times 20$
$KE = 10J$

WORKSHEET – 4

LEVEL – I

1. B	2. C	3. A	4. B	5. A	6. C
7. A	8. A				

LEVEL – II

1. C	2. D	3. B	4. C	5. C	6. D
7. C	8. A	9. D	10. C	11. A	12. C

13. B 14. C

LEVEL – III

15. A, B	16. C	17. B	18. B	19. A-Q, B–P, S, C-P, D-R	
2O. B, C	21. A, B, C	22. A	23. B	24. A	25. A, B, C

SOLUTIONS
LEVEL – II

1. $F = kx$
$x_1 = 20$ cm
$x_2 = 22$ cm
$x = x_2 - x_1$
$x = 22 - 20$
$x = 2$ cm
$k = 5d_{yne}$ cm^{-1}
$F = 5 \times 2$
$F = 10$erg

3. $m = 40$ kg
$E = 980$ joule
$PE = mgh$
$980 = 40 \times 9.8 \times h$
$h = \dfrac{980}{40 \times 9.8}$
$h = 2.5m$

4. $PE = mgh$
$PE = 1000 \times 9.8 \times 100$
$PE = 980000$ J
$PE = 9.8 \times 10^5$ J

7. $w = \frac{1}{2}k(x_2^2 - {}^{2}_{1})$
$w = \dfrac{1}{2} \times 8(15^2 - 5^2)$
$w = \dfrac{1}{2} \times 8(225 - 25)$
$w = \dfrac{1}{2} \times 8(200)$
$w = 800$
$w = 800 \times \dfrac{1}{100}$
$w = 8J$

8. $v = 10m \times 10m \times 10m$
$v = 1000m^3$
$m = d \times v$
$m = 10^3 \times 10^3$
$m = 10^6$
$PE = mgh$
$PE = d \times v \times g \times h$
$PE = 10^3 \times 10^3 \times 10 \times 5$
(Here height = 5 m from the midpoint of tank)
$PE = 5 \times 10^7$ joule

9. $u = 18$ km/hr
$u = 18 \times \frac{5}{18}$ m/s
$u = 5$ m/s
$v = 72$ km/hr
$v = 72 \times \frac{5}{18}$ m/s
$v = 20$ m/s
$m = 2$ kg
$w = \frac{1}{2} \times 2(v^2 - u^2)$ $w = (400 - 25)$
$w = 375J$

10. PE to stretch through x is
$\frac{1}{2}kx^2 = 100J$
To stretch through additional distance' x '
$= \frac{1}{2}k[(2s)^2 - s^2]$
$= 3 \times \frac{1}{2} \times ks^2$
3×100
$PE = 300J$

11. $K = 10N/m$
$PE = \frac{1}{2}k(x_2^2 - x_1^2)$
$PE = \frac{1}{2} \times 10((0.25)^2 - (0.2)^2)$
$PE = \frac{1}{2} \times 10(0.0625 - 0.04)$
$PE = \frac{1}{2} \times 10(0.0225)$
$PE = 0.1125J$

12. $E = 1J$
$x = 20$ cm $\rightarrow x = \frac{20}{100}m$
$k = ?$
$E = \frac{1}{2}kx^2$
$1 = \frac{1}{2} \times k\left(\frac{2}{10} \times \frac{2}{10}\right)$
$100 = 2k$
$k = 50N/m$

13. $E = 50x^2$
$E = \frac{1}{2}kx^2$
$50 = \frac{1}{2}k$
$k = 100Nm$

14. $m = 4$ kg
$p = 8$ kg ms^{-1}
$F = 0.2N$ $a = \frac{F}{m}$

$t = 10\text{sec}\ a = \frac{0.2}{40} = 0.05$
$v = u + at$
$v = 2 + 0.05 \times 10$
$v = 2.5$
$KE = \frac{1}{2}m(v^2 - u^2)$
$KE = \frac{1}{2} \times 4(6.25 - 4)$
$KE = 4.5J$

WORKSHEET – 5

LEVEL – I

1. B	2. B	3. C	4. A	5. B	6. C
7. C	8. D	9. A	10. B		

LEVEL – II

1. B	2. D	3. D	4. B	5. C	6. B
7. B	8. B	9. C	10. C	11. B	12. D
13. C	14. B				

LEVEL – III

15. A, B	16. A-P, B-Q,C-R,D-S,T	17. A	18. C	19. A
20. A	21. C	22. B	23. A-T, B-P, C-Q, D-R	24. C
25. B	26. C			

SOLUTIONS

LEVEL – II

1. $m_1 = 3$
$m_2 = 1$
$E_1 = 1$
$E_2 = 3$
$KE = \frac{1}{2}mv^2$
$v^2 = \frac{2 \times KE}{m}$
$v_1 : v_2 = 1 : 3$

2. $p = 20{,}000$ dyne sec
$KE = 200erg$
$KE = \frac{p^2}{2m}$
$m = \frac{p^2}{2KE}$
$m = \frac{20{,}000 \times 20{,}000}{2 \times 200}$
$m = 10^6$ g
$m = 10^6 \times 10^{-3}$ kg
$m = 10^3$ kg

3. $E_1 = 2$
$E_2 = 1$
$P_1 = 1$
$P_2 = 2$
$m = \frac{P^2}{2E}$
$\frac{m_1}{m_2} = \frac{\frac{p_1^2}{2E_1}}{\frac{P_2^2}{2E_2}}$
$\frac{m_1}{m_2} = \frac{\frac{(1)^2}{2}}{\frac{2^2}{1}}$
$\frac{m_1}{m_2} = \frac{\frac{1}{2}}{\frac{4}{1}}$
$m_1 : m_2 = 1:8$

5. $m = 6$ kg
$s = 8m$
$t = 10\text{sec}$
$F = m\left(\frac{v}{t}\right)$ $KE = \frac{1}{2}mv^2$
$KE = \frac{1}{2} \times 6\left(\frac{s}{t}\right)^2$
$KE = \frac{1}{2} \times 6\left(\frac{8}{10}\right)^2$
$KE = 1.92J$

6. Given,
Initial speed of ball $u = 0$ m/s
Final speed of ball $v = 10$ m/s
$w = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$
$w = \frac{1}{2} \times 0.25(10^2 - 0^2)$
$w = 12.5J$

7. $m = 60$ kg
$E = 750J$
$E = \frac{1}{2}mv^2$
$v = \sqrt{\frac{2E}{m}}$
$v = \sqrt{\frac{2 \times 750}{60}}$
$v = 5$ m/s

8. Loss of energy = KE − PE
Loss of energy $= \frac{1}{2}mv^2 - mgh$
$15 = \text{m}\left[\frac{\text{v}^2}{2} - \text{gh}\right]$

$15 = 3\left[\frac{10^2}{2} - 10 \times \mathrm{h}\right]$
$5 = 50 - 10h$
$10h = 45$
$\mathrm{h} = 4.5$ m

9. $u = 200$ m/s
$s_1 = 5$ cm
$s_2 = 20$ cm
$v^2 - u^2 = 2as$
$u^2 \alpha s$
$\frac{u_1^2}{u_2^2} = \frac{s_1}{s_2}$
$\frac{(200)^2}{u_2^2} = \frac{5 \times 10^{-2}}{20 \times 10^{-2}}$
$u_2^2 = 160000$
$u_2 = 400$ m/s

10. $m = 10g$
$m = 0.01$ kg
$u = 200$ m/s
$v = 300$ m/s
$v^2 - u^2 = 2as$
$(300)^2 - (200)^2 = 2a \times \frac{20}{100}$ $-50000 = \frac{40a}{100}$
$a = -125000$ m/s^2
$F = ma$
$F = 125000 \times \frac{1}{100}$
$F = 1250N$

11. Case I:
$u = 10$ m/s
$v = 20$ m/s
$w_1 = \frac{1}{2}m(v^2 - u^2)$
$w_1 = \frac{1}{2}m(400 - 100)$
$w_1 = \frac{1}{2}m(300) \rightarrow (1)$
Case II:
$u = 0$ m/s
$v = 10$ m/s
$w_2 = \frac{1}{2}m(v^2 - u^2)$
$w_2 = \frac{1}{2}m(100 - 0)$
$w_2 = \frac{1}{2}m(100) \rightarrow (2)$
$\frac{w_1}{w_2} = \frac{1/2m(300)}{1/2m(100)}$
$3 = \frac{w_1}{w_2}$
$w_1 = 3w_2$

12. $v_1 = v$
$s_1 = x$
$v^2 - u^2 = 2as$
$v^2 - u^2 = 2\frac{F}{m}s$
$0^2 - u^2 = 2\frac{F}{m}x \rightarrow (1)$
$v_1 = 2v$
$s_1 = x$
$m = 2m$
$0^2 - (2u)^2 = \frac{2F^1}{2m}x \rightarrow$
Equation (1) & (2)
$\frac{-u^2}{-4u^2} = \frac{2\frac{F}{m}x}{\frac{2F^1}{2m}x}$
$\frac{1}{4} = \frac{F}{F^1/2}$
$F^1 = 8F$

13. Mass of bullet m = 20 g = 0.02 kg Initial velocity of bullet u = 500 m/s Final velocity of bullet v = 400 m/s
$w = \frac{1}{2}m(v^2 - u^2)$
$w = \frac{1}{2}(0.02)((400)^2 - (500)^2)$
$w = -900J$
Work done by the bullet, $w_{\text{bulle}} = -w_{\text{tree}}$
Or $w_{\text{bullet}} = -(-900J)$
w_{bullet} = 900 J

14. m = 4 kg
Increase in height of body h = $(10 - 5)$
h = 5 m
increase in PE
$\Delta PE = mgh$
$\triangle PE = 4 \times 10 \times 5$
$PE = 200J$

LEVEL – III

15. $KE = \frac{1}{2}mv^2 \rightarrow (1)$
$m \rightarrow \frac{m}{4}$
$v \rightarrow 2v$
$KE = \frac{1}{2}\frac{m}{4} \times (2v)^2$
$KE = \frac{1}{2}\frac{m}{4} \times 4v^2$
$KE = \frac{1}{2}mv^2 \rightarrow (2)$
From (1) & (2) it remains same

20. $PE = mgh$

$PE_1 = mgh \rightarrow (1)$

$PE_2 = mg(3h) \rightarrow (2)$

$\frac{PE_1}{PE_2} = \frac{mgh}{3mgh}$

$PE_1 : PE_2 = 1:3$

25. Work done = change in KE

$w = \frac{1}{2} 0.01[(1000)^2 - (500)^2]$

$w = 3750$ J

26. work done = change in KE

Initial KE = 0, final KE = $\frac{1}{2}mv^2$

v → final speed

$50000 = \frac{1}{2} \times m \times v^2$

$50000 = \frac{1}{2} \times 3 \times 10^7 v^2$

$v^2 = \frac{1}{100} = 0.1$ m/s

WORKSHEET – 6

LEVEL – I

1. B	2. B	3. D	4. A	5. B	6. A
7. B	8. A				

LEVEL – II

1. B	2. A	3. B	4. C	5. C	6. A
7. A	8. C	9. B	10. A	11. A	12. A
13. C	14. D	15. B	16. D		

LEVEL – III

17.A, B	18. C	19. C	20. B	21. A-Q,R; B-R; C-S,T; D-P
22. A, B, C, D	23. C	24. A	25. D	26. B 27. D
28. B	29. A, D	30. A-Q,R; B-R; C-S,T; D-P		

SOLUTIONS

LEVEL – II

7. $E = \frac{1}{2}mv^2 - mgh$

$E = \frac{1}{2} \times 5 \times 20^2 - 5 \times 9.8 \times 12$

$E = 412J$

8. $PE = mgh$

$h \rightarrow \frac{4h}{5}$

$PE = \frac{mgh}{5} \rightarrow (1)$

$KE = \frac{1}{2}mv^2$

$v = \sqrt{2g \times \frac{h}{5}}$

$KE = \frac{1}{2}m\left(\frac{2gh}{5}\right)$

$KE = \frac{mgh}{5} \rightarrow (2)$

$\frac{KE}{PE} = \frac{\frac{mgh}{5}}{\frac{4mgh}{5}}$

$KE : PE = 1 : 4$

12. $PE + KE = TE$

$2 + 3 = TE$

$TE = 5$

$TE = \frac{1}{2}mv^2 = 5J$

If velocity doubles

$TE = \frac{1}{2}m(2v)^2$

$TE = \frac{1}{2}m4v^2$

$TE = 4 \times \frac{1}{2}mv^2$

$TE = 4 \times 5$

$TE = 20J$

$TE = KE + PE$

$20 = KE + 2$

$KE = 18J$

$\frac{PE}{KE} = \frac{2}{18}$

$PE : KE = 1 : 9$

13. $TE = PE + KE$

$mgh = 3.6 + 6.2$

$0.1 \times 9.8h = 9.8$

$h = 10m$

14. $\frac{KE}{PE} = \frac{\frac{1}{2}mv^2}{mgh}$

$\frac{KE}{PE} = \frac{\frac{1}{2} \times 40 \times 1}{20 \times 10 \times 1.2}$

$\frac{KE}{PE} = 1 : 12$

26. $w = \frac{1}{4}KE(\text{loss } 25\%)$

$F \times x = \frac{1}{4}KE$

$F.s = KE$

$\frac{x}{s} = \frac{1}{4}$

$s = 4x$

LEVEL – III

27. Since the forces are acted upon simultaneously by two forces 4 N&3 N at right angles, net force = $\sqrt{4^2 + 3^2}$
$= 5N$
The mass is given as
$10 \text{ kg} \rightarrow a = \frac{F}{m}$
$a = \frac{5}{10} = 0.5 \text{ m/s}^2$
$KE = \frac{1}{2}mv^2$
$KE = \frac{1}{2}m(at)^2$
$KE = \frac{1}{2} \times 10 \times (0.5 \times 10)^2$
$KE = 125J$

28. Kinetic energy required by body = total work done
$= F \times s - \mu mgs$
$= 25 \times 10 - 0.2 \times 5 \times 10 \times 10$
$= 250 - 100$
$KE = 150$ joule

7. HEAT

WORKSHEET – 1

LEVEL – I

1.C 2. B 3. A 4. A 5. B 6. A 7. C
8. B 9. C

LEVEL – II

1.C 2. B 3. C 4. A 5. C 6. A 7. D
8. A 9. D 10. A 11. C 12. B 13. C 14. A
15. A 16. A 17. C 18. A

LEVEL – III

19. A, B 20. A, B 21. C
22. A 23. A 24. D 25. A-Q, B-P, C-R, D-S 26. A
27. A, B, C, D 28. 1000 29. A

WORKSHEET – 2

LEVEL – I

1. A 2. B 3. D 4. D 5. B 6. B 7. C 8. B
9. A 10. C

LEVEL – II

1. A 2. C 3. B 4. A 5. C 6. B 7. B 8. A
9. B 10. B 11. A 12. D 13. C 14. D 15. A 16. B
17. 3 18. A 19. B

LEVEL – III

20 C, D 21. A, B, D
22. A 23. A 24. B 25. C 26. A-P; B-Q; C-P; D-S,T 27. D
28. C 29. D 30. 7

SOLUTIONS

LEVEL – II

1. $\frac{C-0}{100-0} = \frac{F-32}{212-32}$
2. $\frac{C}{100} = \frac{F-32}{180}; \frac{40}{100} = \frac{F-32}{180}; F = 104^0F$ increase in temperature
$= 104°F - 32°F = 72°F$
6. $\frac{60-(-10)}{x-(-10)} = \frac{50-0}{100-0}$

$$\frac{70}{x+10} = \frac{50}{100}$$

$140 = x + 10$

$x = 130°C$

7. Conceptual

8. $\frac{C-0}{100-0} = \frac{59-5}{95-5}$

$\frac{C}{100} = \frac{54}{90}$

$C = 60°C$

9. $F = 3C$

$\frac{C}{5} = \frac{3C-32}{9}$

$9C = 15C - 160$

$15C - 9C = 160$

$6C = 160$

$C = \frac{80}{3}$ °C

10. $K = x; F = x; \frac{F-32}{9} = \frac{K-273}{5}; \frac{x-32}{9} = \frac{x-273}{5}$

$5x - 160 = 9x - 2457$

$4x = 2297; x = 574.25$

11. F = x

K = 2x

$\frac{x-32}{9} = \frac{2x-273}{5}$

$5x - 160 = 18x - 2457$

$2297 = 13x$

x = 176.9°F

C = n

K = 2n

$\frac{C}{5} = \frac{K-273}{5}$

$n = 2x - 273$

$273 = 2x - x$

x = 273°C

12.

X

Y

80

120 → Divisions

LUP20

30

9–?

120

$\frac{120-30}{120} = \frac{x-20}{80}$

$\frac{90}{3} = \frac{x-20}{2}$

$x = 60 + 20$

x = 80

13.

X

JFP → 120°C

$$\frac{50}{\frac{50-40}{120-40}} = \frac{y-(-30)}{130-(-30)}$$

$$\frac{10}{80} = \frac{y+30}{160}$$

$$20 = y + 30$$

$$y = -30 + 20$$

$$= -10°C$$

	X	Y
UFP →	120°C	−30°C
LFP →	140°C	130°

14. UFP → 120°F

LFP → 140°F

body temperature 40 °C

$$\frac{C-0}{100-0} = \frac{F-30°}{210-30°}$$

$$\Rightarrow \frac{C}{100} = \frac{F-30°}{180}$$

$$\Rightarrow \frac{40}{100} = \frac{F-30}{180}$$

$$18(4) = F - 32$$

$$72 + 32 = F$$

$$F = 104°F$$

15. F = R = x

$$\frac{F-32}{9} = \frac{R}{4}$$

$$\frac{x-32}{9} = \frac{x}{4}$$

$$4x - 128 = 9x$$

$$x = \frac{-128}{5} = -25.6$$

16. C = x

F = 2x

$$\frac{C}{5} = \frac{F-32}{9}$$

$$\frac{x}{5} = \frac{2x-32}{9}$$

$$9x = 10x - 160$$

$$x = 160$$

$$\therefore C = x = 160°C$$

$$F = 2x = 320°C$$

17. Faulty $x = 39, LFP = 5, UFP = 95$

$$\frac{C-0}{100-0} = \frac{x-5}{95-5}$$

$\frac{C}{100} = \frac{39-5}{90}$

$C = \frac{340}{9}$

$C = 37.78°C$

18. Faulty reading $= 60°C$

LFP $= -3$, UFP $= 103$

Coreect reading $C = ?$

$\frac{C}{100} = \frac{60+3}{103+3}$

$\frac{C}{100} = \frac{63}{106}$

$C = \frac{6300}{106}$

$C = 59.43°C$

19. $\frac{C}{100} = \frac{F-32}{180}; \frac{C}{5} = \frac{113-32}{9};$

$C = 45°C$

$+1°C$

LEVEL – III

20. C, D

21. $C = 50°$

$\frac{C}{5} = \frac{F-32}{9}; \frac{50}{5} = \frac{F-32}{9};$

$F = 90 + 32; F = 122°F$

$C = 50°; \frac{C}{5} = \frac{R}{4}; \frac{50}{5} = \frac{R}{4}; R = 40°R$

$\frac{C}{5} = \frac{K-273}{5}; \frac{50}{5} = \frac{K-273}{5};$

$K = 50 + 273; K = 323K$

22. A

23. A

24. B

25. C

26. A-P; B-Q; C-P; D-S, T

27. $\frac{C}{5} = \frac{100-273}{5}; C = -173°C$

28. $\frac{C}{5} = \frac{K-273}{5}; \frac{27}{5} = \frac{K-273}{5};$

$K = 27 + 273; K = 300K$

29. $\frac{F-32}{9} = \frac{R}{4}; \frac{100-32}{9} = \frac{R}{4}; \frac{68\times4}{9} = R$

$R = 30.2^0R$

30. $\frac{C}{5} = \frac{F-32}{9}; \frac{C}{5} = \frac{95-32}{9};$

$\frac{C}{5} = \frac{63}{9} = 7 \times 5^0C$

8. CIRCULAR MOTION

WORKSHEET – 1

LEVEL – I

1. B	2. D	3. A	4. C	5. D	6. A	7. D
8. C						

LEVEL – II

1. C	2. A	3. A	4. B	5. A	6. B	7. B
8. A	9. B	10. C	11. C	12. A	13. D	14. A
15. B	16.D					

LEVEL – III

17. A, B, C, D	18. D	19. A	20. A		
21. A-Q, B-P, C-R, D-S	22. A	23. A	24. A	25. D	26. A, B
27. A-Q; B-P; C-S; D-R	28. B				

SOLUTIONS

LEVEL – II

1. C
2. A
3. A
4. $r = 6400m, \omega = ?$

   $\omega = \frac{2\pi}{T}\ T = 24 \times 60 \times 60S$

   $= \frac{2 \times 3.14}{24 \times 60 \times 60}$

   $\omega = 7.3 \times 10^{-5}\text{rads}^{-1}$
5. $\omega = \frac{2\pi}{t}$

   $\omega = \frac{2\pi}{3600} = \frac{\pi}{1800}\frac{\text{rad}}{\text{s}}$

   6. $\frac{180}{3} = 60°$
7. $\frac{180}{4} = 45°$
8. $f = 20$ in 10sec

   $f = \frac{20}{10} = 2\text{rev/s}$

   $\omega = 2\pi f$

   $= 2\pi \times 2$

   $= 4\pi\text{rad/s}$
9. $\omega = \frac{2\pi}{T}$

T → Time period

$T = 24 \times hr$

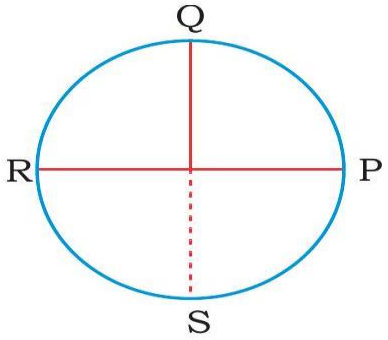
$= 24 \times 60 \times 60s$

$= 24 \times 3600s$

$= 86{,}400s$

$\omega = \frac{2\pi}{86400} = \frac{\pi}{43{,}200}\text{rad/s}$

10. $\frac{1}{4}$ th of circle when it travel from P to Q_o



11. $\frac{4\pi}{27} = \frac{4\times180}{27} = \frac{80}{3}$

12. $\omega = 20\text{rad/s}$
$d = 40\text{ cm}$
$= 40 \times 10^{-2}\text{ cm}$
$= 4 \times 10^{-1}\text{ m}$
$= \frac{4}{10} = 0.4m$
$d = 0.4m$
$r = \frac{d}{2} = \frac{0.4}{2} = 0.2$
$v = r\omega$
$= 0.2 \times 20$
$= \frac{2}{10} \times 20$
$v = 4m/s$

13. $v = 10\text{ m/s}$
$r = 45\text{ cm} = 45 \times 10^{-2}\text{ m}$
$\omega = ?$
$v = r\omega$
$\omega = \frac{v}{r} = \frac{10}{45 \times 10^{-2}}$
$= \frac{10 \times 10^2}{45}$
$= \frac{10^3}{45} = \frac{1000}{45}$
$= 22.2\text{rad/s}$

14. car scooter
$v_2 = v\ v_2 = r$
$r_2 = 2r\ r_1 = r$
$\omega_2 = ?\ \omega_1 = ?$
$\frac{\omega_1}{\omega_2} = ?\ \omega = \frac{v}{r}$
$\frac{\omega_1}{\omega_2} = \frac{v_1/r_1}{v_2/r_2} = \frac{v_1}{r_1} \times \frac{r_2}{v_2}$
$\frac{\omega_1}{\omega_2} = \frac{v_1}{v_2} \times \frac{r_2}{r_1} = \frac{v}{v} \times \frac{2r}{r} = \frac{2}{1}$
$\frac{\omega_2}{\omega_1} = \frac{1}{2} = 1:2$

15. $\theta = \frac{\pi}{2}$
$t = 5s$

$\omega = \frac{\theta}{t} = \frac{\pi}{10}$ rad/s

16. $a = \frac{v^2}{r} = \frac{8\times8}{8} = 8\ m/s^2$

LEVEL – III

18 Given that radius of circular path = 7 m
The speed of the particle = 7 ms^{-1}
As particle revolves through quarter circle from A to B its m
$AB = \sqrt{7^2 + 7^2} = 7\sqrt{2}\ m/s^2$
Time taken by the particle to complete
$1rev = \frac{2\pi r}{v} = \frac{2\pi\times7}{7} = 2\pi$sec ond
So, time taken to revolve through
quarter circle $t = \frac{2\pi}{4} = \frac{\pi}{2}$ sec ond
Average velocity $= \frac{\text{Total Displacement}}{\text{Time Taken}}$
$= \frac{7\sqrt{2}}{\pi/2} = \frac{14\sqrt{2}}{\pi} = \frac{14\times7\sqrt{2}}{22}$
$= \frac{98\sqrt{2}}{22} = \frac{98\times1.414}{22}$
$= 6.3\ m/s^2$

19. Magnitude of acceleration
$|a| = \frac{v^2}{r} = \frac{7\times7}{7} = 7\ ms^{-2}$

20. Magnitude of average acceleration = ? $|a_{avg}|$ = ?
$|a_{avg}| = \frac{\text{Change in velocity}}{\text{Time taken travelling quarter circle}}$ magnitude of change in velocity
$= |Vf - Vi| = 2V\sin\frac{\theta}{2}$
$= 2\times7\times\frac{1}{\sqrt{2}} = 7\sqrt{2}$
time taken $= \frac{2\pi r}{7} = \frac{2\pi7}{7} = 2\pi s$
to complete 1 rev time taken to complete quarter circle
$= \frac{2\pi}{4} = \frac{\pi}{2}S$
$|a_{avg}| = \frac{7\sqrt{2}}{\pi/2} = \frac{14\sqrt{2}}{22/7} = \frac{14\sqrt{2}\times7}{22} = \frac{138.52}{22}$
$= 6.298 = 6.3\ m/s^2$

23. Angular velocity
$\omega = \frac{n(2\Pi)}{t} = \frac{(7)(3.14)}{100} = \frac{7\times6.88}{100} \oplus .44\frac{\text{rad}}{\text{s}}$

24. $v = rw = (12)(0.44) = 5.3$ cm/s

25. $a = r\alpha \Rightarrow a = r(0) = 0$

26. $\omega = \frac{2\pi}{t} = \frac{2\pi}{60} = \frac{\pi}{30}\frac{\text{rad}}{\text{s}}$

27. $a_c = \frac{v^2}{r}$
$3g = \frac{\left(108\times\frac{5}{18}\right)^2}{r}$

$30 = \frac{30^2}{r}$

$r = 30m$

WORKSHEET – 2

LEVEL – I

1. B	2. D	3. A	4. A	5. C	6. D	7. B
8. C						

LEVEL – II

1. B	2. B	3. C	4. B	5. C	6. B	7. B
8. D	9. B	10. C	11. D	12. D	13. D	14. C
15. D	16. B					

LEVEL – III

17. A, B, D	18. A	19. A	20. A	21. A-R; B-P; C-Q; D-S,T
22. A, B, C		23. A	24. D	25. C

SOLUTIONS

LEVEL – II

1. $r = 0.5$ m
   $\omega = 4\text{rad/s}$
   $v = r\omega = 0.5 \times 4$
   $= \frac{5}{10} \times 4 = 2$ m/s
2. $r = 2$ m
   $a_t = 10$ m/s^2
   $a_t = r\alpha$
   $\alpha = \frac{at}{r} = 10/2 = 5\text{rad/s}^2$
3. $\omega_1 = 0, \omega_2$
   $= 45, t = 15$
   $\alpha = ?$
   $\omega_2 - \omega_1 = \alpha t$
   $\alpha = \frac{\omega_2 - \omega_1}{t}$
   $= \frac{45 - 0}{15} = \frac{45}{15}$
   $= 3\text{rad/s}^2$
4. $r = 5m$
   $a_t = 50$
   $\alpha = ?$
   $a_t = r\alpha$
   $\alpha = \frac{a_t}{r} = \frac{50}{5} = 10\text{rad/s}^2$
5. $\omega_1 = 300\text{rpm} = 300 \times \frac{2\pi}{60}\text{rad/s};$
   $\omega_1 = 10\pi; \omega_2 = 30\pi;$

$\alpha = \frac{30\pi - 10\pi}{5} = 4\pi \text{rad/s}^2$

$\omega_2^2 - \omega_1^2 = 2\alpha(2\pi n)$

$900\pi^2 - 100\pi^2 = 2(4\pi)(2\pi n)$

$= 800\pi^2 = 16\pi^2 \times n; n = 50$

6. For hours hand, $\frac{1}{12}$ rev/hour

For minutes hand, $\frac{1}{1}$ rev / hour

ratio $= \frac{\frac{1}{12}}{1} = \frac{1}{12} = 1:12$

7. $\omega = \frac{v}{r}; \omega \alpha v$

$\frac{\omega_1}{\omega_2} = \frac{v_1}{v_2} = \frac{3}{6} = \frac{1}{2}$

8.

$r_1^1 = \frac{r}{2}, \omega = ?, V$ is constant

$\omega^1 = \frac{V}{r^1} = \frac{v}{r/2} = \frac{2v}{r} = 2\omega$

$\therefore$ increase by 2 times

9. $\omega_1 = 300\text{rpm}$

$300 \times \frac{2\pi}{60}$

$\omega_1 = 10\pi \text{rad/s}$

$\omega_2 = 0, \alpha = -4\text{rad/s}^2$

$t = ?$

$t = \frac{\omega_{2-}\omega_1}{\alpha}$

$t = \frac{0 - 10\pi}{-4} = \frac{-10\pi}{4} = 2.5\frac{\pi}{s}$

10. $\omega_0 = 1200\text{rpm} = 1200 \times \frac{2\pi}{60} = 40\pi \text{rads}^{-1}$

$\alpha = -4\text{rad/s}^2, \omega = 0$

$\omega_2^2 - \omega_1^2 = 2\alpha\theta$

$0^2 - (40\pi)^2 = 2 \times -4 \times \theta$

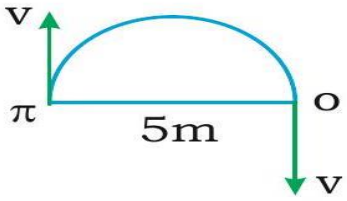
$1600\pi^2 = 8\theta$

$\theta = \frac{1600\pi^2}{8}$

$\theta = 200\pi^2 (\therefore \theta = 2\pi N)$

rev $= \frac{200\pi^2}{2\pi} = 100\pi = 100 \times 3.14 = 314$

11



$r = 5m, V = 5\ \text{ms}^{-1}, a = ?$

Half revolution $= \frac{2\pi r}{2} = \pi r$

$T = \frac{s}{v} = \frac{\pi r}{v} = \frac{\pi \times 5}{5} = \pi s$

acceleration,

$a = \frac{\Delta V}{\Delta t} = \frac{2v}{\pi} = \frac{2 \times 5}{\pi} = \frac{10}{\pi} \text{m/s}^2$

12. $r = 1$ cm

$\Delta V = ?$

$\Delta V = 2V\sin\frac{\theta}{2} = 2V\sin 45° = 2V\left(\frac{1}{\sqrt{2}}\right)$

But $V = \frac{\pi/2}{15} = \frac{\pi}{30}$

$\therefore \Delta V = 2 \times \frac{\pi}{30} \times \frac{1}{\sqrt{2}}$

$\therefore \Delta V = \frac{\pi\sqrt{2}}{30}$ cm/s

13. $r_{1=10^{12}m}, r_2 = 10^{13}m, T_1.T_2 = ?$

[ω_1, r are constant]

$V = r\omega = \frac{2\pi r}{T}$

$\therefore \frac{T_1}{T_2} = \frac{r_1}{r_2} = \frac{10^{12}}{10^{13}} = \frac{1}{10}$

$\therefore T_1 : T_2 = 1 : 10$

14. radial acceleration = Tangential acceleration

$a_t = a_r$

$\alpha r = \omega^2 r$

$\alpha = \omega^2$

$\omega = \alpha t - - - (1)$

from $\alpha = \omega^2$

$\alpha = (\alpha t)^2 \Rightarrow \alpha = \alpha^2 t^2$

$t^2 = \frac{1}{\alpha}$

$t = \frac{1}{\sqrt{\alpha}}$

15. $t_1 = t_2$

$\frac{2\pi}{\omega_1} = \frac{2\pi}{\omega_2}$

$\frac{\omega_1}{\omega_2} = 1$

16. Thrust = weight - centripetal force

Thrust = $mg - mv^2/r$.

LEVEL – III

23. $V = r\omega \Rightarrow v = 10(7)(2\pi)$

$= 140\pi\text{cm/s} = 1.4\pi\text{m/s}$

25. $\text{a} = \frac{\omega^2}{\text{r}} = \left(\frac{22}{44}(2\Pi)\right)^2$

$= \Pi^2$ m/s^2 towards centre

WORKSHEET – 3

LEVEL – I

1. B 2. A 3. C 4. B 5. B 6. D 7. A

8. C 9. C

LEVEL – II

1. D	2. C	3. C	4. C	5. D	6. B	7. D
8. B	9. B	10. C	11. A	12. A	13. B	14. C
15. D	16. D	17. C	18. B			

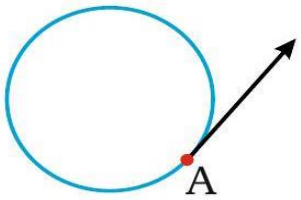
LEVEL – III

19. A	20. A-R,T; B-S,T; C-Q; D-P		21. A	22. B	23. D
24. D	25. A	26. A, B, C, D	27. A	28. B	

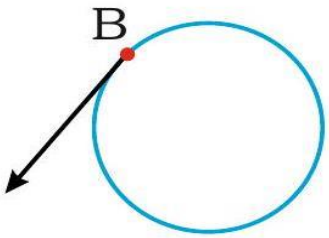
SOLUTIONS

LEVEL – II

1. $v = \frac{2\pi r}{t} = \frac{2\pi \times 4}{2} = 4\pi \; a = \frac{v^2}{r} = \frac{(4\pi)^2}{4} = \frac{16\pi^2}{4} = 4\pi^2$
2. C
3. C
4. C
5. D
6. B
7. When ever an object is going in a circular path the direction of the velocity of the object is going to act tangential of the circle.
   So at the point A.



8. When ever an object is going in a circular path the direction of the velocity of the object is going to act tangential of the circle.
   So at the point B.



9. Equal in magnitude but opposite direction.
10. $m_1 = m_2, r_1 : r_2 = 1:2$
    $F_1 = F_2, V_1 : V_2 = ? \; F = \frac{mv^2}{r}$
    $\frac{v^2}{r} = \text{cons tan } t$
    $\therefore \frac{V_1}{V_2} = \sqrt{\frac{r_1}{r_2}} = \sqrt{\frac{1}{2}}$
    $\therefore V_1 : V_2 = 1:\sqrt{2}$
11. A
12. $a = 2.5 \text{ ms}^{-2}, m = 10 \text{ kg}, r = 10 \text{ m}$

$a = \frac{V^2}{r}$

$\Rightarrow V = \sqrt{ar} = \sqrt{2.5 \times 10}$

$= \sqrt{25} = 5$ m/s

$\therefore V = 5$ m/s

13. $V = 1080\text{Kmph}, a = 3g, r = ?$

$= 60 \times 5$ m/s, $a = 3 \times 10 = 30$ m/s^2

$a = \frac{v^2}{r} \Rightarrow r = \frac{v^2}{a} = \frac{60 \times 5 \times 60 \times 5}{30}$

$\therefore r = 3000m$

$r = 3$ km

14. $a = \frac{v^2}{r} = r\omega^2;$

$r \times \left(\frac{2\pi}{T}\right)^2 \left[\omega = \frac{2\pi}{T}\right]; \frac{r \times 4\pi^2}{T^2} = \frac{4\pi^2 r}{T^2}$

15. $F = \frac{mv^2}{r}$

$m = \frac{F \times r}{v^2} = \frac{5 \times 1}{5 \times 5} = 0.2$

$\text{mass}(m) = 0.2$ kg

16. $f_{CP} = \frac{mv^2}{r} = \frac{1500 \times 81}{300} = 405$ N

17. $v = \frac{2\pi r}{t}, F_{cp} = \frac{mv^2}{R} = \frac{0.4 \times 7.85 \times 7.85}{5}$

$v = \frac{2\pi r}{t} = \frac{2 \times 3.14 \times 5}{4} = 7.82$ m/s

$= 4.93$ N

18. $v = 3 \times 2\pi r = 6\pi r$ [1 cycle is $2\pi r$]

$F_{cp} = \frac{mv^2}{r} = \frac{36 \times \pi^2 \times r^2}{r}$

$= 36 \times 10 \times 2 = 720$ N

LEVEL – III

21. $F_c = \frac{mv^2}{r} \Rightarrow \frac{m^2v^2}{mr} = \frac{P^2}{mr}$

22. $F_c = \frac{mv^2}{r}$

$40 = \frac{(10)v^2}{4} \Rightarrow v^2 = \frac{160}{10} \Rightarrow v = 4$ m/s

23. $F_c = mru^2 \Rightarrow F_c = (20)(5)(10)^2$

$F_c = 10^4$ N

25.

$F_c = \frac{mv^2}{r} = \frac{2}{2}\frac{mv^2}{r} = \frac{2E}{r} \left(\therefore \frac{mv^2}{R} = KE\right)$

27. m = 0.1 kg

r = 3 m, v = 30 m/s

$\text{F} = \text{mv}^2/\text{r} = \frac{(0.1)30 \times 30}{3} = 30$ N

WORKSHEET-4

LEVEL – I

1. D	2. A	3. D	4. A	5. A	6. B	7. B
8. D						

LEVEL – II

1. C	2. B	3. B	4. B	5. A	6. D	7. B
8. B	9. B	10. C	11. C	12. A	13. D	14. A
15. C	16. B	17. A, B, C, D				

LEVEL – III

18. D	19. C	20. A			
21. C	22. A-R; B-P; C-Q; D-S	23. B	24. A	25. A	26. A

SOLUTIONS

LEVEL – II

1. C

$V = 36\text{Kmph}, r = 20\text{ m}$

$= 36 \times \frac{5}{18} = 10\text{ m/s}\ \theta = ?$

$\theta = \tan^{-1}\left(\frac{V^2}{rg}\right)$

$\theta = \tan^{-1}\left(\frac{V^2}{rg}\right) = \tan^{-1}\left(\frac{10 \times 10}{20 \times 10}\right) = \tan^{-1}\left(\frac{1}{2}\right)$

$\therefore \theta = \tan^{-1}\frac{1}{2}$

3. $V = 10\text{ m/s}, r = 50\text{ m}, g = 10\text{ m/s}^2, \theta = ?$

$\theta = \tan^{-1}\left(\frac{V^2}{rg}\right) = \tan^{-1}\left[\frac{10 \times 10}{50 \times 10}\right]$

$\theta = \tan^{-1}\left(\frac{1}{5}\right)$

4. $v = \sqrt{\mu_s rg} = \sqrt{0.4 \times 30 \times 10}$

$= \sqrt{120} = 11\text{ m/s}$

8. $V = 60\text{ km/h}\ r = 0.1\text{ km}\ \theta = ?$

$= \frac{50}{3} m/s, V = 0.1 \times 1000 = 10^2\text{ m}$

$\theta = \tan^{-1}\left(\frac{V^2}{rg}\right)$

$= \tan^{-1}\left(\frac{50 \times 50}{3 \times 3 \times 10^2 \times 10}\right)$

$= \tan^{-1}\left(\frac{5}{18}\right)$

$\therefore \theta = \tan^{-1}\left(\frac{5}{18}\right)$

9. $V = 72\text{kmph},\ r = 20\text{ m}$

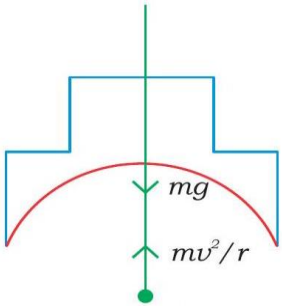
$= 200\text{ m/s}\ g = 10\text{ m/s}^2$

$\theta = ?$

$\theta = \tan^{-1}\left(\frac{V^2}{rg}\right) = \tan^{-1}\left[\frac{20 \times 20}{20 \times 10}\right]$

$\therefore \theta = \tan^{-1}(2)$

10.



$mg - N = \frac{mv^2}{r}$

$N = mg - \frac{mv^2}{r}$

Centrifugal force balance to resultant force.

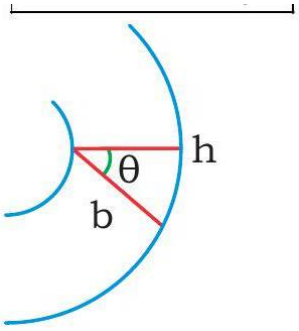
11. $r = 10m, \theta = 10°, V = ?$

$\tan\theta = \frac{V^2}{rg} V^2 = rg\tan\theta$

$= 10 \times 10\tan 10° = 100 \times 0.1763$

$= 17.63 \therefore V = 4.2$ m/s

12.



From fig, $\tan\theta = \frac{h}{b}$

But $\tan\theta = \frac{V^2}{rg}$

$\frac{h}{b} = \frac{V^2}{rg}$

$\therefore h = \frac{V^2 b}{rg}$

13. $\tan\theta = \frac{v^2}{rg}$

$\frac{2}{10} = \frac{v^2}{50 \times 10}$

$v^2 = 100$

v = 10 m/s

14. $\tan\theta = \frac{v^2}{rg}$

$v_1 = v$

$v_2 = v + 0.1v;\ \frac{v^2}{r_1} = \frac{v_2^2}{r^2}$

$r_2 = \frac{v_2^2 r_1}{v_1^2} = \frac{(1.1v_1)^2}{v_1^2} \times 20$

$r_2 = 24.2m$

15. $T = 2\pi\sqrt{\frac{l}{g}} = 2 \times 3.14\sqrt{\frac{1}{10}}$

$2 \times 3.14\sqrt{0.1} = 2 \times 3.14 \times 0.31 = 1.98$

16. $v = 108$ km/h, $r = 120$

$108 \times \frac{5}{18} = 30$ m/s

$\tan\theta = \frac{v^2}{rg} = \frac{30 \times 30}{120 \times 10} = \frac{3}{4}; \theta = 37°$

LEVEL – III

23. $\tan\theta = \frac{v^2}{rg}$

$\tan 37° = \frac{v^2}{10 \times 10}; \frac{3}{4} = \frac{v^2}{100}$

$; v^2 = \frac{3}{4} \times 100; v = \sqrt{\frac{3}{4}} \times 100$

$= \sqrt{3} \times \frac{10}{2} = 5\sqrt{3}$ m/s

24. $\theta = \tan^{-1}\left(\frac{v^2}{rg}\right)$

$= \tan^{-1}\left(\frac{\left(120 \times \frac{5}{18}\right)^2}{400 \times 9.8}\right)$

$= \tan^{-1}(0.2834)$

25. $F = \frac{mv^2}{r}$

$= \frac{10^5 \times 20^2}{150} = 2.67 \times 10^5$ N

26. $\theta = \tan^{-1}\left(\frac{v^2}{rg}\right) = \tan^{-1}\left(\frac{400}{150 \times 9.8}\right)$

$= \tan^{-1}(0.2721)$

9. OPTICS

WORKSHEET – 1

LEVEL – I

1.C 2. A 3. C 4. A 5. B 6. A 7. A
8. B 9. D

LEVEL – II

1.B 2. A 3. C 4. D 5. A 6. A 7. B
8. D 9. A 10. B 11. C 12. D 13. A 14. D

LEVEL – III

15. B, D 16. B, C, D 17. A 18. A 19. 20. C
21. A- Q; B-R; C-P; D-S 22. A 23. A 24. A-P,T; B-S; C-P; D-S
25. A, B, C 26. A 27. 3×10^8 28. A-R, B-S, C-P, D-Q

WORKSHEET – 2

LEVEL – I

1.B 2. C 3. C 4. A, D 5. A 6. C 7. A
8. C

LEVEL – II

1.D 2. D 3. C 4. A 5. C 6. A 7. A
8. D 9. D 10. B 11. D 12. B 13. A 14. A

LEVEL – III

15. A, B, C 16. B 17. C 18. A 19. A 20. A
21. A-R; B-S; C-P; D-Q 22. B 23. A 24. A, B 25. A

WORKSHEET – 3

LEVEL – I

1.B 2. C 3. D 4. D 5. A 6. C 7. B
8. B

LEVEL – II

1.B 2. D 3. B 4. B 5. B 6. B 7. A
8.A 9. A 10. A 11. A 12. B 13. C 14. A, C
15. B 16. A 17. B 18. A 19. A-Q; B-P; C-S; D-T 20. A, B
21. A, B, D

WORKSHEET – 4

LEVEL – I

1.A 2. A 3. A 4. B 5. A 6. A 7. D
8. C 9. A

LEVEL – II

1.C 2. B 3. B 4. A 5. B 6. A 7. C
8. B 9. C 10. C 11. A 12. A 13. A 14. B

15. A

LEVEL – III

16. A, B, C 17. A 18. D 19. C 20. D
21. A-Q; B-P; C-S; D-Q 22. B 23. C 24. C 25. A, B, C 26. 90°

---

SOLUTIONS

LEVEL – II

2. Given incident ray and reflected ray are at right angle to each other. Then
$i + r = 90° \Rightarrow 2r = 90° \Rightarrow r = 45°$
3. Glancing angle $= 40°$
$g + r = 90° \Rightarrow 40° + r = 90° \Rightarrow r = 50°$
4. For normal incidence, $i = 0$ and $r = 0$
5. $\delta = 60° \Rightarrow \delta = 180 - 2i$
$60 = 180 - 2i \Rightarrow 2i = 120 \Rightarrow i = 60°$
6. $i = 60° \Rightarrow i + g = 90 \Rightarrow g = 30°$
7. $i + r = 90° \Rightarrow 2i = 90° \Rightarrow i = 45°$
$\delta = 180 - 2i = 180 - 90 = 90°$
8. $\delta = 180° - 2i \Rightarrow 3i = 180° \Rightarrow i = 60°$
then, $i + g = 90° \Rightarrow g = 30°$
9. $g + i = 90° \Rightarrow 2i + i = 90° \Rightarrow i = 30°$
$\delta = 180 - 2i = 180° - 60° = 120°$
10. $i + g = 90° \Rightarrow i + 30° = 90° \Rightarrow i = 60°$
$\delta = 180° - 2i = 180° - 120° = 60°$
$\delta = 180° - 120° = 60°$
11. $\delta = 180° - 2i \Rightarrow 2i = 180 - 2i$
$\Rightarrow 4i = 180 \Rightarrow i = 45°$
12. For normal incidence
$i = 0°, r = 0° \Rightarrow \delta = 180 - 2i \Rightarrow \delta = 180°$
13. $i = 30°$
$\delta = 180 - 2i \Rightarrow \delta = 180° - 60° \Rightarrow \delta = 120°$
14. $\delta = 180° - 2i = 180° - 160° = 20°$
15. $i + g = 90° \Rightarrow 60 + g = 90° \Rightarrow g = 30°$

LEVEL – III

16. $\delta = 180 - 2i \Rightarrow \delta = 180 - 90° \Rightarrow \delta = 90°$
17. $\delta = 180° - 2i \Rightarrow 100i = 180° - 2i$
$2i = 80° \Rightarrow i = 40°$
25. $g = 35°$
$i + g = 90° \Rightarrow i = 55°, r = 55°$

WORKSHEET – 5

LEVEL – I

1. A 2. D 3. A 4. A 5. C 6. A 7. A
8. C

LEVEL – II

1.B	2. B	3. C	4. A	5. B	6. A	7. D
8. A	9. A	10. A	11. A	12. C	13. D	14. A
15. A						

LEVEL – III

16. D	17. D	18. D	19. A, B			
20. A-R; B-S,T; C-P; D-Q		21. 10 CM	22. B	23. A	24. A	
25. A	26. B	27. B	28. A			

SOLUTIONS

LEVEL – I

1. 3feet. Becasue minimum size of mirror is equal to half of the height of person.
2. Minimum height of the mirror is equal to half of the height of object.
3. Size of mirror is equal to half of height of person.
4. Size of person standing infront of mirror = size of image formed inside the mirror.
5. 10 cm. Size of person standing infront of mirror = size of image formed inside the mirror.
6. When converging beam incident on a plane mirror, real image is formed

---

WORKSHEET – 6

LEVEL – I

1.A	2. A	3. A	4. C	5. C	6. C	7. A

LEVEL – II

1.A	2. B	3. C	4. B	5. B	6. B	7. D
8. A	9. D	10. D	11. D	12. A-R; B-P; C-Q; D-T		13.2cm
14. A	15. B	16. A	17. A	18. B	19. D	
20. A, B, C, B		21. C	22. A	23. A	24. B	25. B
26. C	27. D					

SOLUTIONS

LEVEL – II

2. Image speed is 2 u. i.e., 10 m/s.
3. When the object moves with speed 10 m/s towards (or away) from the plane mirror then image also moves towards (or away) with speed 10 m/s.
4. When the object moves with speed 20 m/s towards (or away) from the plane mirror then image also moves towards (or away) with speed 20 m/s.
5. Given, $i = 90°, d = 180° - 2\theta = 180° - 2i$
   $d = 180° - 2 \times 90° = 180° - 180° = 0°$
9. $n = \frac{360°}{\theta} - 1 = \frac{360°}{30°} - 1 = 12 - 1 = 11$

10. a) $n = \frac{360^\circ}{60^\circ} - 1 = 6 - 1 = 5$

b) $n = \frac{360^\circ}{45^\circ} - 1 = 8 - 1 = 7$

c) $n = \frac{360^\circ}{90^\circ} - 1 = 4 - 1 = 3$

d) $n = \frac{360^\circ}{20^\circ} - 1 = 18 - 1 = 17$

13. Same displacement that is 2 cm.
14. Reflected ray rotates by 20°
15. Reflected ray rotates by 80°
16. $n = \frac{360^\circ}{\theta} - 1 = \frac{360}{90} - 1 = 4 - 1 = 3$
17. dor $\delta = 180^\circ - 2i = 180^\circ - 2 \times 60^\circ = 60^\circ$
18. $n = \frac{360^\circ}{36} - 1 = 10 - 1 = 9$
19. Reflected ray is rotated by an angle 40°
20. Reflected ray is rotated by an angle 60°
21. $d = 180^\circ - 2i^\circ = 180^\circ - 2 \times 40 = 100^\circ$

WORKSHEET – 7

LEVEL – I

1.B	2. C	3. A	4. D	5. B	6. D	7. B
8. C	9. D					

LEVEL – II

1.A	2. A	3. B	4. D	5. D	6. A	7. B
8.C	9. A	10. A	11. A	12. B	13. C	14. C
15. C	16.C	17. D	18. B			

LEVEL – III

19. A, C	20. A	21. A	22.D	23. C		
24. A-S; B-T; C-P; D-Q		25. 2.5 CM	26. A	27.D	28. A	29. A
30. C						

SOLUTIONS

LEVEL – II

2. $f = \frac{R}{2} = 12.5$ cm
3. $R = 2f = 32$ cm
4. $f = 25, R = 2f = 50$ cm
5. $R = 18$ cm, $f = \frac{R}{2} = 9$ cm
6. $f = 10$ cm, $R = 2f = 20$ cm
7. $f_1 = 30$ cm, $R_1 = 2f_1 = 60$ cm

$f_2 = 10$ cm, $R_2 = 2f_2 = 20$ cm

8. $R = 40$ cm, $f = \frac{R}{2} = 20$ cm
9. $R = 45$ cm, $f = \frac{R}{2} = \frac{45}{2} = 22.5$ cm

10. $R = 60$ cm, $f = \frac{R}{2} = \frac{60}{2} = 30$ cm
11. f = 15 cm, R = 2f = 2(15) = 30 cm
12. $R = 24$ cm, $f = \frac{R}{2} = \frac{24}{2} = 12$ cm
13. $f = 36$ cm, $R = 2f = 2(36) = 72$ cm
14. $f = 1.2$ cm, $R = 2f = 2(1.2) = 2.4$ m
15. R = 1 m = 100 cm
    $f = \frac{R}{2} = 50$ cm
18. $R = 20$ cm, $f = \frac{R}{2} = \frac{20}{2} = 10$ cm

LEVEL – III

19. $R = 5$ cm, $f = \frac{R}{2} = \frac{5}{2} = 2.5$ cm
20. $R = 0.4$ cm, $f = \frac{R}{2} = \frac{0.4}{2} = 0.2$ cm
21. $R = 1m, f = \frac{R}{2} = \frac{1}{2} = 0.5m$
22. $R = 12$ cm, $f = \frac{R}{2} = 6$ cm
23. $f = x$, $R = 2f = 2x$

∴ Sum of focal length and Radius of curvature is $(x + 2x) = 3x$

WORKSHEET – 8

LEVEL – I

1.A 2. A 3. B 4. A 5. B 6. A 7.A
8. D 9. A

LEVEL – II

1.A 2. C 3. A 4.C 5. C 6.D 7. B
8.C 9. B 10.B 11.D 12. B 13. A
14. C

LEVEL – III

15.A, B, C 16. A-P,Q,T; B-P,Q,R; C-S; D-P,Q 17. B 18. A
19.B 20. A 21. D 22.A 23. B

WORKSHEET – 9

LEVEL – I

1. A 2. B 3. B 4. B 5. A 6. B 7. B
8. A

LEVEL – II

1. A 2. B 3. A 4. B 5. C 6. A 7. A
8. A 9. A 10. A 11. C 12. D 13. A 14. B
15.D 16. C 17.A 18. B 19. A 20. D 21. A, B
22. A-Q; B-S; C-P; D-R 23. A 24. B 25. A 26. C 27. B
28. B

SOLUTIONS

LEVEL – II

4. $u = -20\ cm, f = -15\ cm$

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$-\frac{1}{15} = \frac{1}{v} - \frac{1}{20}$$

$$-\frac{1}{15} + \frac{1}{20} = \frac{1}{V}$$

$$\frac{-4+3}{60} = \frac{1}{v}$$

$$\frac{-1}{60} = \frac{1}{V}$$

$$V = -60\ cm$$

5. $\frac{1}{f} = \frac{1}{v} + \frac{1}{u} \Rightarrow \frac{1}{15} = \frac{1}{v} + \frac{1}{18} \Rightarrow v = \frac{90}{11}\ cm$

6. $\frac{1}{f} = \frac{1}{v} + \frac{1}{u} \Rightarrow \frac{-1}{5} = \frac{1}{v} - \frac{1}{10} \Rightarrow v = -10\ cm$

7. $\frac{1}{f} = \frac{1}{v} + \frac{1}{u} \Rightarrow \frac{-1}{10} = \frac{1}{v} - \frac{1}{5} \Rightarrow v = 10\ cm$

$$\frac{hi}{ho} = \frac{-v}{u} \Rightarrow \frac{hi}{5} = \frac{-10}{-5} \Rightarrow hi = 10\ cm$$

10. $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$

$$\frac{1}{-15} = \frac{1}{v} + \frac{1}{45}$$

$$\therefore v = -22.5\ cm$$

11. $u = -10\ cm, f = -15\ cm$

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$\frac{-1}{15} = \frac{-1}{-10} + \left(\frac{1}{v}\right)$$

$$\therefore v = -40\ cm$$

13. $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$

$$\frac{2}{-15} = \frac{1}{v} - \left(\frac{1}{10}\right)$$

$$\therefore v = -30\ cm$$

$$m = \frac{-v}{u} = \frac{-(-30)}{-10}$$

$$m = -3$$

14. $\frac{1}{f} = \frac{-1}{u} + \frac{1}{v}$

$$\frac{1}{v} = \frac{1}{f} + \frac{1}{u} = \frac{1}{f} - \frac{1}{f} = 0 \Rightarrow v = \infty$$

LEVEL – III

15. When object is at C. Image is also at C. minimum seperation between object and its real image is zero.
16. For both convex and concave mirrors.
17. A concave mirror can form a real as well as a virtual image for real object.

18. $\frac{1}{f}=\frac{1}{v}+\frac{1}{u}\Rightarrow\frac{1}{10}=\frac{1}{v}-\frac{1}{20}\Rightarrow v=\frac{20}{3}$ cm

19. $m=\frac{-v}{u}=\frac{-20/3}{-20}=1/3$

20. $\frac{h_i}{h_o}=\frac{-v}{u}\Rightarrow\frac{h_i}{0.5}=\frac{1}{3}\Rightarrow h_i=0.167$ m

21. $\frac{1}{\mathrm{f}}=\frac{1}{\mathrm{v}}+\frac{1}{\mathrm{u}}\Rightarrow\frac{-1}{20}=\frac{1}{\mathrm{v}}-\frac{1}{-30}\Rightarrow \mathrm{v}=-60$ cm

22. $\frac{1}{f}=\frac{1}{v}+\frac{1}{u}\Rightarrow\frac{-1}{10}=\frac{1}{v}\frac{-1}{5}\Rightarrow v=10$ cm

$$m=\frac{-v}{u}=\frac{-10}{-5}=2$$